DANNY'S RACE CAR

ELECTRICAL HARNESS / REV J

L94 / LS3 | Nexus R3 | Delta ABS Gen 2 | IC-7 | 15-button CAN keypad

DESIGN REVIEW DRAFT — OPEN THIS ROOT FILE IN KICAD

02 / BATTERY, ISOLATION AND STARTER

File: 02_Power.kicad_sch

03 / NEXUS R3 PIN ALLOCATION

File: 03_R3_Pin_Allocation.kicad_sch

04 / ENGINE POWER, INJECTION AND IGNITION

File: 04_Engine_Power.kicad_sch

05 / ENGINE SENSORS AND WIDEBAND

File: 05_Engine_Sensors.kicad_sch

06 / DRIVE-BY-WIRE

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08 / DELTA ABS GEN 2 MAIN CONNECTOR

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09 / ABS SENSORS, CONTROLS AND PUBLIC CAN

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17 / WOLFBOX G900 PRO CAMERA POWER

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18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS

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19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS

File: 19_Stock_Engine_Harnesses.kicad_sch

WHAT IS VERIFIED

R3 and PD16 connector functions; Delta supplied pin allocations; owner equipment choices. Circuit assignments are a draft.

WHAT REMAINS OPEN

TBD devices, selected ratings, sensor cavity mapping, ABS compatibility, isolation, CAN configuration and physical harness routing.

DRAFT — NOT A BUILD RELEASE

Danny's Race Car

Sheet: /

File: Race_Car_RevJ.kicad_sch

Title: Race car electrical harness — system index

Size: A3	Date: 2026-10-27	**Rev: J — REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 1/19

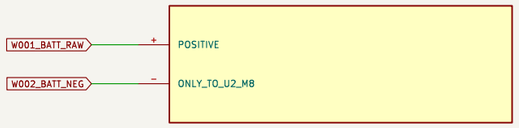
02 / BATTERY, ISOLATION AND STARTER

CARTEK GT is the master isolator; K1 removed. F0 feeds TB2 directly. Returns are downstream of isolation.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

BT1 | BRAILLE G30H

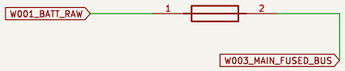
Battery



PHYSICAL ROUTE: battery-side BATT_RAW block -> F2 -> M1 battery B+ stud -> F3 -> G1 B+.

F0 | MAIN BUS FUSE / SELECT

MAIN BUS FUSE / SELECT



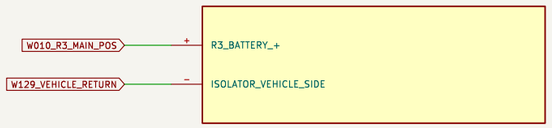
F1 | R3 FEED / SELECT

Engine controller supply fuse



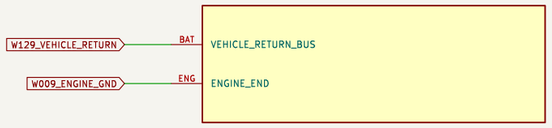
J1 | R3 MAIN SURLOK

Engine controller power connector



J2 | GROUND STRAP / ENGINE

Engine ground strap



U2 isolates NEGATIVE. Positive starter/alternator cables remain battery-connected when OFF: insulate, route and protect them.

GT M6_IGN -> S1 -> R3 A13 is the engine-stop control. F0 -> TB2 is permanently battery-positive when connected.

F5 protects the relay-contact/solenoid branch; value TBD. Estimated 30A solenoid pull-in is NOT a measurement. No motor current through U3.

K1 omitted by owner. Verify R3 ignition-OFF stops injection/ignition/fuel and PD16 loads, including CAN-loss/shutdown timing.

No wire except U2 M8 may return to BT1 negative. R3/PD16 and every load use VEHICLE_RETURN; verify with all service cables.

W011: one shared start/charge trunk. F3 branches at M1 BATTERY B+ stud, never SOL or motor-side stud. Size/protection TBD.

U3 / 24059 BULKHEAD RELAY WIRING

Large PWR <- F5/2; large OUT -> M1 SOL.

Small CMD <- R3 A18/HBO5 (+12V ON).

Small GND -> VEHICLE_RETURN (not case).

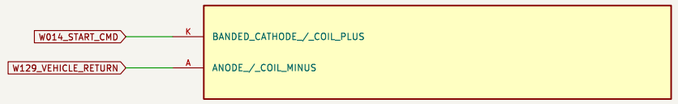
D60 band/K -> CMD; unbanded/A -> GND.

PWR/OUT/CMD/GND are installer tags.

12V coil: 17.5 ohm / ~0.69A calculated.

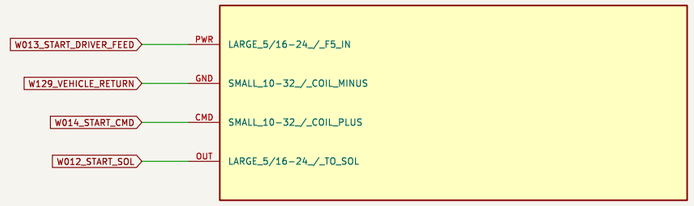
D60 | 1N5408 / BAND TO CMD

Starter-relay coil suppression diode



U3 | COLE HERSEE 24059 / 12V / 85A

Cole Hersee 24059 starter relay



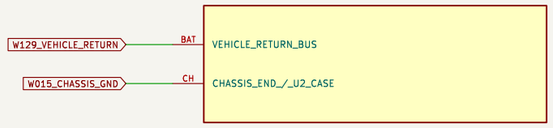
F5 | START DRIVER FUSE / SELECT

START DRIVER FUSE / SELECT



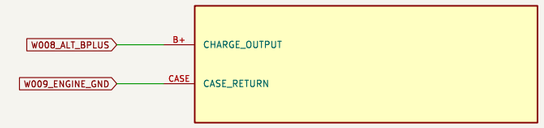
J3 | GROUND STRAP / CHASSIS

Chassis ground strap



G1 | 79376G / ONE-WIRE

Alternator



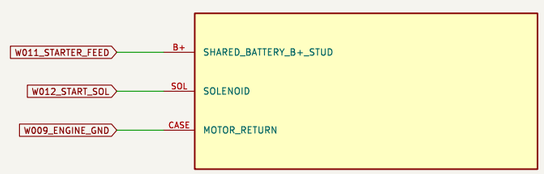
F3 | ALT BRANCH FUSE / AT STARTER

ALT BRANCH FUSE / AT STARTER



M1 | VW HIGH-TORQUE STARTER

Starter motor



F2 | SHARED CABLE FUSE / AT BATTERY

Shared starter / charging cable fuse



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Danny's Race Car | Harness design

Sheet: /02 / BATTERY, ISOLATION AND STARTER/

File: 02_Power.kicad_sch

Title: 02 / BATTERY, ISOLATION AND STARTER

Size: A3	Date: 2026-10-27	**Rev: J – REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 2/19

03 / NEXUS R3 PIN ALLOCATION

ONE R3 with owner-selected HT-183200 flying-lead harness. Supplied controller leads need routing and free-end termination.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J10 | R3 A / AMP34 KEY1

Engine controller — connector A

W016_INJ_1	1	INJECTOR_1
W017_INJ_2	2	INJECTOR_2
W018_INJ_3	3	INJECTOR_3
W019_INJ_4	4	INJECTOR_4
W020_INJ_5	5	INJECTOR_5
W021_INJ_6	6	INJECTOR_6
W022_INJ_7	7	INJECTOR_7
W023_INJ_8	8	INJECTOR_8
	9	DPO1_/_SPARE
W025_R3_SHIELD_RETURN	10	GROUND_OUT_/_SHIELDS
W026_R3_CAN_RETURN	11	GROUND_OUT_/_CAN
	12	DPO2_/_SPARE
W028_R3_WAKE	13	IGN_SWITCH_INPUT
	14	DPO3_/_SPARE
	15	DPO4_/_SPARE
	16	DPO5_/_SPARE
	17	DPO6_/_SPARE
W014_START_CMD	18	HBO5_/_START_CMD
W029_DBW_MA	19	HBO1_/_DBW_MOTOR_A
W030_DBW_MB	20	HBO2_/_DBW_MOTOR_B
W130_ABS_IGN_SOURCE	21	HBO3_/_ABS_IGNITION
W032_COCKPIT_12V	22	HBO4_/_COCKPIT_12V
W033_CAN1_H	23	CAN1_HIGH
W034_CAN1_L	24	CAN1_LOW
W131_PD16_WAKE	25	HBO6_/_PD16_WAKE
	26	IGN_SOURCE_/_UNUSED
W036_IGN_1	27	IGNITION_1
W037_IGN_2	28	IGNITION_2
W038_IGN_3	29	IGNITION_3
W039_IGN_4	30	IGNITION_4
W040_IGN_5	31	IGNITION_5
W041_IGN_6	32	IGNITION_6
W042_IGN_7	33	IGNITION_7
W043_IGN_8	34	IGNITION_8

J11 | R3 C / AMP34 KEY2

Engine controller — connector C

W044_CRANK_SIG	1	TRIGGER_+
	2	TRIGGER_-_/_HOLD
W045_CAM_SIG	3	HOME_+
	4	HOME_-_/_HOLD
W137_TC_TRIM_SIG	5	SPI1_/_TC_ROTARY_-_PULLUP_OFF
W047_BRAKE_REQUEST	6	SPI2_/_BRAKE_REQUEST
	7	SPI3_/_SPARE
	8	SPI4_/_SPARE
	9	SENSOR_8V_/_SPARE
W048_APP1	10	AVI1_/_APP1
W049_APP2	11	AVI2_/_APP2
W050_TPS1	12	AVI3_/_TPS1
W051_TPS2	13	AVI4_/_TPS2
W052_OIL_PRESS	14	AVI5_/_OIL_PRESS
W053_FUEL_PRESS	15	AVI6_/_FUEL_PRESS
W054_COOLANT_TEMP	16	AVI7_/_COOLANT_TEMP
W055_AIR_TEMP	17	AVI8_/_AIR_TEMP
	18	AVI9_/_SPARE_-_ONBOARD_MAP
	19	SPI5_/_SPARE
	20	SPI6_/_SPARE
W057_CAN2_H	21	CAN2_HIGH
W058_CAN2_L	22	CAN2_LOW
W059_KNOCK1	23	KNOCK1
W060_KNOCK2	24	KNOCK2
W061_R3_REF5	25	SENSOR_5V_/_200mA
W062_R3_SG	26	SIGNAL_GROUND
W063_GEAR_SIG	27	AVI10_/_GEAR
W123_SHIFT_FORCE_SIG	28	AVI11_/_SHIFT_FORCE
W064_WBO_HTR_PLUS	29	WBI1_/_HTR_PLUS
W065_WBO_INPUT	30	WBI1_/_INPUT
W066_WBO_PUMP	31	WBI1_/_PUMP
W067_WBO_NERNST	32	WBI1_/_NERNST
W068_WBO_HTR_MINUS	33	WBI1_/_HTR_MINUS
W069_WBO_CAL	34	WBI1_/_CAL

One wideband: onboard WBI1 / C29-C34. No external WB controller. Confirm sensor type/adapter; configure NSP.

A10/A11 are ground OUTPUTS: do not bond them directly to battery, engine or chassis. C26 is sensor return only.

DBW HBO1/2; HBO3 = ABS ignition; HBO4 = cockpit; HBO6 = PD16 wake. Separate protected feeds; verify each load.

HT-183200 supplies labeled flying leads, CAN plugs and wideband plug. Keep shielding; verify revision, reach and sensor adapters.

SPI3-6 remain spare; AVI11/C28 = HGT. A26 is NC: CARTEK U2 M6_IGN is the only source for S1 / R3 A13 (sheet 18).

DRAFT — NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /03 / NEXUS R3 PIN ALLOCATION/
File: 03_R3_Pin_Allocation.kicad_sch

Title: 03 / NEXUS R3 PIN ALLOCATION

Size: A3	Date: 2026-10-27	**Rev: J — REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 3/19

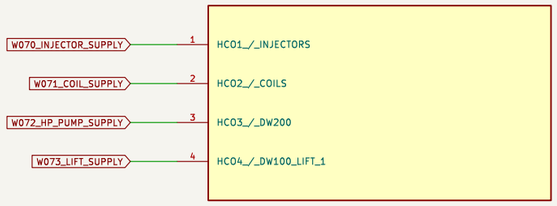
04 / ENGINE POWER, INJECTION AND IGNITION

2011 L94 with LS3 cam/intake/throttle body | Donor connectors reused only after condition and cavity checks.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J12 | R3 E / DTP4

Haltech R3 — integrated PDM outputs



ELECTRONIC OUTPUT PROTECTION / NO DOWNSTREAM BANK FUSES

HCO1 –> all injector +12V branches; HCO2 –> all coil +12V branches.

Set fuse current / startup time / delay / retries in NSP.

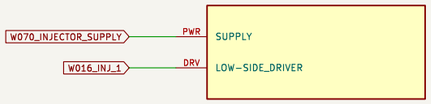
Each channel senses the SUM of its connected loads.

All branch wires and contacts must tolerate the programmed trip curve.

25A is output capacity, NOT a selected protection setting.

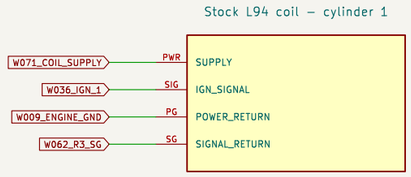
INJ1 | INJECTOR / CYL 1

LS3 injector — cylinder 1



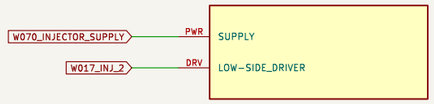
IC1 | COIL / CYL 1

Stock L94 coil — cylinder 1



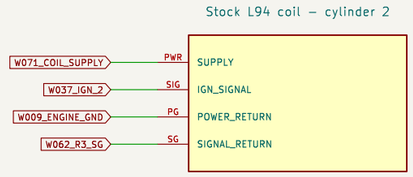
INJ2 | INJECTOR / CYL 2

LS3 injector — cylinder 2



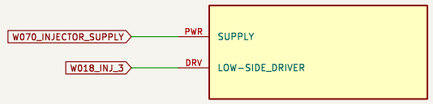
IC2 | COIL / CYL 2

Stock L94 coil — cylinder 2



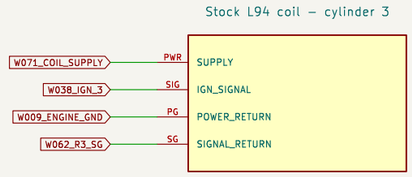
INJ3 | INJECTOR / CYL 3

LS3 injector — cylinder 3



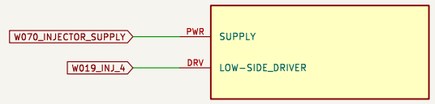
IC3 | COIL / CYL 3

Stock L94 coil — cylinder 3



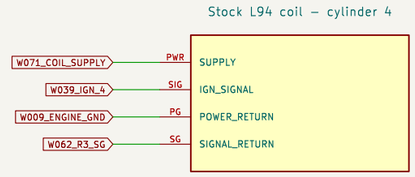
INJ4 | INJECTOR / CYL 4

LS3 injector — cylinder 4



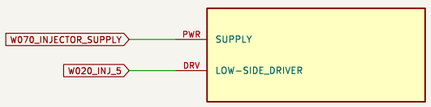
IC4 | COIL / CYL 4

Stock L94 coil — cylinder 4



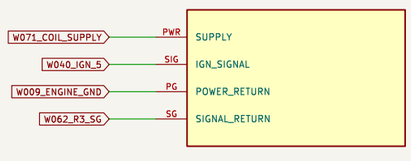
INJ5 | INJECTOR / CYL 5

LS3 injector — cylinder 5



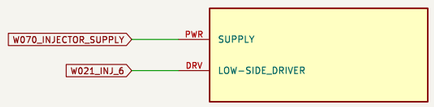
IC5 | COIL / CYL 5

Stock L94 coil — cylinder 5



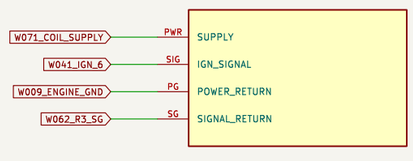
INJ6 | INJECTOR / CYL 6

LS3 injector — cylinder 6



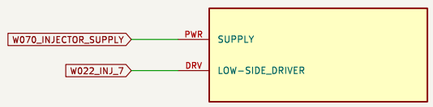
IC6 | COIL / CYL 6

Stock L94 coil — cylinder 6



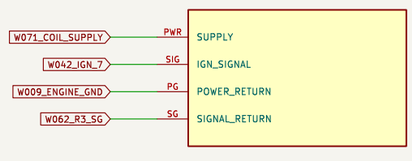
INJ7 | INJECTOR / CYL 7

LS3 injector — cylinder 7



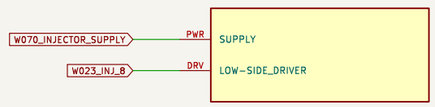
IC7 | COIL / CYL 7

Stock L94 coil — cylinder 7



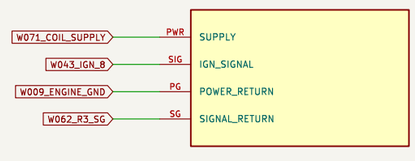
INJ8 | INJECTOR / CYL 8

LS3 injector — cylinder 8



IC8 | COIL / CYL 8

Stock L94 coil — cylinder 8



HOLD P02: HCO1/2 must protect retained OEM wire/contact branches. Stock L94 coil and injector wiring is retained; see sheet 19.

PWR/DRV/SIG/PG/SG remain functional device IDs. Stock bank connector reference cavities are on sheet 19; inspect donor before wiring.

IGN and INJ channel number = cylinder number; firing order is an ECU configuration item, not a reason to reorder wires.

Do not merge HCO1/HCO2 through donor shared feeds. Coil low-reference grounding remains HOLD pending exact coil/R3 evidence.

DRAFT — NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /04 / ENGINE POWER, INJECTION AND IGNITION/

File: 04_Engine_Power.kicad_sch

Title: 04 / ENGINE POWER, INJECTION AND IGNITION

Size: A3	Date: 2026-10-27	**Rev: J — REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 4/19

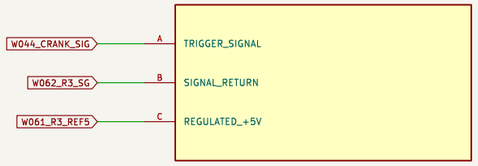
05 / ENGINE SENSORS AND WIDEBAND

R3 analog allocation is on sheet 03. Sensor model, reference voltage and calibration must match before connection.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

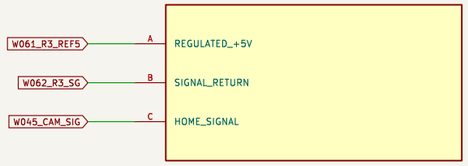
B1 | CRANK / GM 12703627

Crankshaft position sensor



B2 | CAM / GM 12591720

Camshaft position sensor



MAP SENSOR IS INSIDE NEXUS R3

Onboard 4-bar absolute MAP selected in NSP.

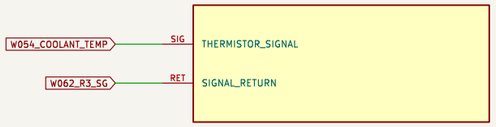
Intake manifold -> pressure hose -> R3 MAP barb.

Pneumatic connection only; no external sensor wires.

AVI9 / C18 spare (onboard MAP). One wideband only.

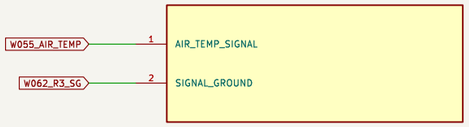
B4 | COOLANT TEMP

Coolant temperature sensor



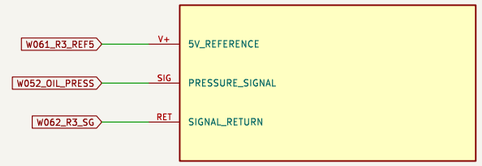
B5 | HT-010206 / INTAKE AIR

Haltech HT-010206 intake air temperature



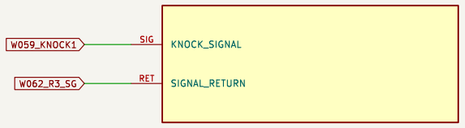
B6 | OIL PRESSURE / TBD

Oil pressure sensor



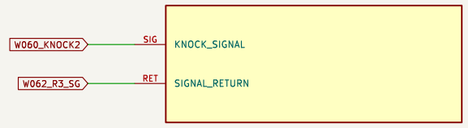
B7 | KNOCK / BANK 1

Knock sensor - bank 1



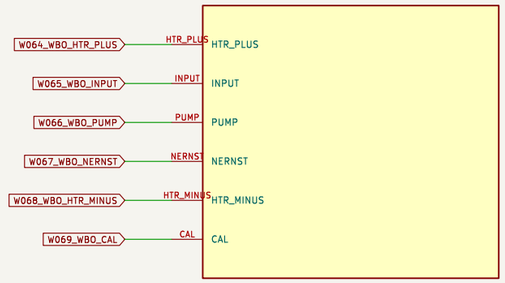
B8 | KNOCK / BANK 2

Knock sensor - bank 2



B9 | ONE WIDEBAND / R3 WBI1

Single wideband O2 / R3 onboard



Crank A -> C1; B -> C26; C -> C25. Cam A -> C25; B -> C26; C -> C3. W078/W079 retired; never reuse.

Shield crank/cam/knock cables; shield/drain to R3 A10 only at ECU end. Sensor return remains C26.

Owner confirms crank GM12703627 and cam GM12591720. Verify molded cavities; C25 200mA includes BOTH sensors.

WBI1 C29-C34: HTR+ / INPUT / PUMP / NERNST / HTR- / CAL. Retain matching sensor/adapter; no separate power/ground.

HT-010206: C17, 1k pull-up, GM Air calibration. O2: select LSU4.9 or NTK in NSP; exact pack/cavities and exhaust location TBD.

DRAFT - NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /05 / ENGINE SENSORS AND WIDEBAND/

File: 05_Engine_Sensors.kicad_sch

Title: 05 / ENGINE SENSORS AND WIDEBAND

Size: A3	Date: 2026-10-27	**Rev: J - REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 5/19

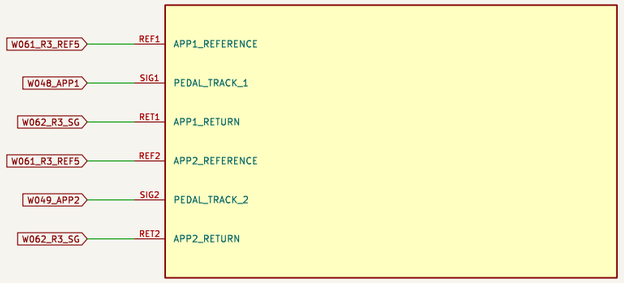
06 / DRIVE-BY-WIRE

Pedal GM 19417903 | Stock LS3 throttle body | Function labels below are NOT physical cavity numbers.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

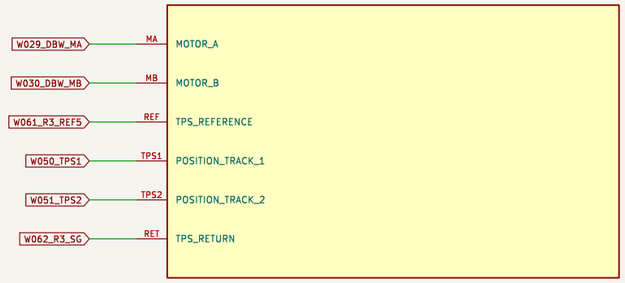
B10 | C6 PEDAL / GM 19417903

Accelerator pedal



M2 | LS3 THROTTLE BODY

Electronic throttle body



S2 REMOVED / NO CLUTCH SWITCH

R3 C5 / SPI1 now serves S50 TC trim; W046 remains retired.

Fresh start request, engine-not-running and bounded crank remain required.

No clutch/neutral switch input; physical master stop remains independent.

HOLD D01: verify all six pedal cavities and all throttle-body cavities, including motor polarity, before terminating.

Pedal pigtail missing: donor connector may be reusable; CPP-LSAPP is a listed candidate. Housing fit does not prove wiring.

APP1/APP2 -> C10/C11; TPS1/TPS2 -> C12/C13. No pull-ups assumed for powered position signals.

Complete Haltech DBW calibration, track correlation, brake override and fault checks before any powered vehicle test.

Owner removes S2 and its two wires. No clutch/neutral start interlock is drawn; SPI1 must not be jumpered.

DRAFT - NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /06 / DRIVE-BY-WIRE/

File: 06_DBW.kicad_sch

Title: 06 / DRIVE-BY-WIRE

Size: A3	Date: 2026-10-27	**Rev: J - REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 6/19

07 / FUEL PUMPS, SENDERS AND COOLING

Two DW100 lift pumps -> surge tank -> DW200 feed pump | PROFORM 67049 kit: two 67048 brushless puller fans; 10 A each; independent on/off feeds.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

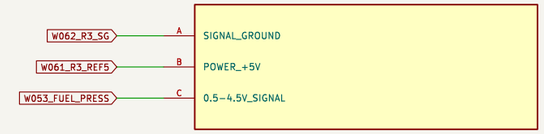
LIFT PUMPS: SEPARATE PROTECTED CHANNELS

Lift 1 = R3 E4; lift 2 = PD16 B4. No F31/F32 inline fuses.

Set each trip curve for its own leads; verify cold/hot startup.

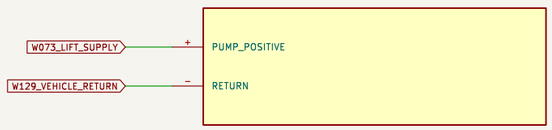
B11 | HT-010910 / 150 PSI FUEL

Haltech HT-010910 fuel pressure



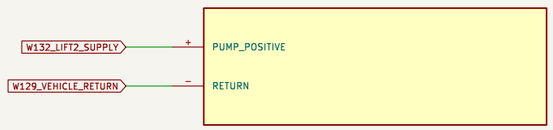
M3 | DW100 / LIFT 1

Fuel lift pump 1



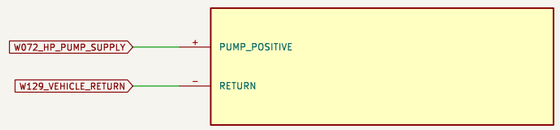
M4 | DW100 / LIFT 2

Fuel lift pump 2



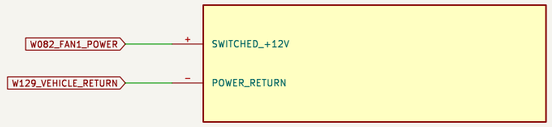
M5 | DW200 / HIGH PRESSURE

High-pressure fuel pump



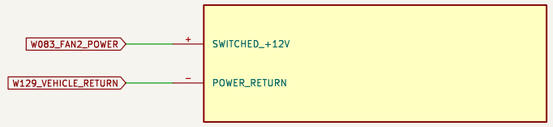
M6 | PROFORM 67048 / 10 A

Radiator fan 1



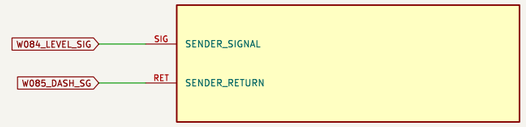
M7 | PROFORM 67048 / 10 A

Radiator fan 2



B12 | FUEL LEVEL / SN38T10

Fuel level sender



HT-010910: A ground / B +5V / C signal -> R3 AVI6 C15. Pull-up OFF; 0.5V = 0 PSI / 4.5V = 150 PSI.

Remote-mount HT-010910 with flexible fuel-rated hose; do not mount directly on rail/block. PROFORM pair: 20 A listed.

Lift pumps now have independent 25A-capable outputs. Select lower trip thresholds from measured load and lead/contact ratings.

Fuel permission requires engine run or timed prime AND valid PD16 communication/output status. Trip/CAN loss -> fuel/engine stop.

Fuel pumps: prime/run logic with engine-stall shutdown; service prime must be momentary/time-limited and master-stop subordinate.

Fans: PD16 B1/B2 supply independent switched +12V (100% when on). No motor-power PWM or external speed-control signal.

Fans: 11.43 x 10.75 x 3.25 in each; pair 22.86 x 10.75 in on 24 x 12 in core. Rear clearance confirmed; verify brackets/shroud.

DRAFT - NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /07 / FUEL PUMPS, SENDERS AND COOLING/

File: 07_Fuel_Cooling.kicad_sch

Title: 07 / FUEL PUMPS, SENDERS AND COOLING

Size: A3	Date: 2026-10-27	**Rev: J - REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 7/19

08 / DELTA ABS GEN 2 MAIN CONNECTOR

Bosch 1928405167 / 38 cavities | Source: owner-supplied SCS Delta Gen 2 Rev.3 allocation.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J20 | DELTA HCU / 38 PIN

Anti-lock brake unit — main connector

W086_ABS_PUMP_POS	1	PUMP_SUPPLY_/_40A_NOTE
	2	FR_SPEED_OUT_/_UNUSED
	3	ABS_ACTIVE_/_UNUSED
W087_ABS_FR_RET	4	FR_SENSOR_RETURN
W088_ABS_SLIP_5V	5	SLIP_5V_REF
W089_ABS_PRESS_5V	6	PRESSURE_5V_REF
W090_ABS_SLIP_RET	7	SLIP_RETURN
W091_ABS_FL_RET	8	FL_SENSOR_RETURN
W092_ABS_SG9	9	SENSOR_RETURN
W093_ABS_WARN_SINK	10	WARNING_LAMP_SINK
	11	SPEEDOMETER_/_UNUSED
	12	RR_SPEED_OUT_/_UNUSED
W094_ABS_MAIN_POS	13	SUPPLY_/_25A_NOTE
W095_ABS_PORT2_L_TBD	14	CAN2_LOW_/_HOLD
W047_BRAKE_REQUEST	15	BRAKE_LIGHT_SINK
	16	RL_SPEED_OUT_/_UNUSED
	17	FL_SPEED_OUT_/_UNUSED
W096_ABS_RL_RET	18	RL_SENSOR_RETURN
W097_ABS_FL_SIG	19	FL_SENSOR_SIGNAL
W098_ABS_PORT1_L_TBD	20	CAN1_LOW_/_HOLD
	21	ANALOG1_/_SPARE
W099_ABS_SLIP_SIG	22	SLIP_ANALOG_SIGNAL
W100_ABS_SG23	23	ANALOG_RETURN
	24	ANALOG2_/_SPARE
W101_ABS_POWER_RETURN	25	PUMP_GROUND
W102_ABS_PORT2_H_TBD	26	CAN2_HIGH_/_HOLD
	27	ANALOG3_/_SPARE
W103_ABS_WAKE	28	IGNITION_WAKE
W104_ABS_RR_RET	29	RR_SENSOR_RETURN
	30	ANALOG4_/_SPARE
W105_ABS_RL_SIG	31	RL_SENSOR_SIGNAL
W106_ABS_RR_SIG	32	RR_SENSOR_SIGNAL
W107_ABS_PORT1_H_TBD	33	CAN1_HIGH_/_HOLD
W108_ABS_FR_SIG	34	FR_SENSOR_SIGNAL
W109_ABS_ENABLE	35	ENABLE_/_GROUND_ON
W110_ABS_PRESS_FRONT	36	FRONT_PRESSURE
W111_ABS_PRESS_REAR	37	REAR_PRESSURE
W101_ABS_POWER_RETURN	38	POWER_GROUND

F41 | ABS PUMP / VALUE TBD

Anti-lock brake pump fuse

W003_MAIN_FUSED_BUS — 1 — 2 — W086_ABS_PUMP_POS

F42 | ABS MAIN / VALUE TBD

Anti-lock brake control fuse

W003_MAIN_FUSED_BUS — 1 — 2 — W094_ABS_MAIN_POS

F43 | ABS WAKE / VALUE TBD

Anti-lock brake wake supply fuse

W130_ABS_IGN_SOURCE — 1 — 2 — W103_ABS_WAKE

J21 | ABS GROUND / HOLD

Anti-lock brake ground connection

W101_ABS_POWER_RETURN	ABS	HARNESS_RING
W129_VEHICLE_RETURN	BAT	PROPOSED_BATTERY_END

Owner retains complete Delta loom and its fuses. Guide: pump40A/solenoid20A/ignition10A; older25A label requires revision check.

J21 records the proposed battery return; HOLD A01: reconcile manual chassis-ground wording with pinout battery-negative wording.

HOLD A02: identify which HCU CAN pair serves public CAN and which serves the supplied IMU/diagnostic harness.

ABS sensor grounds remain within ABS. Main connector unused output pins have NC marks; public-interface detail follows.

Finished Delta branches and IMU/private CAN are supplied, not wires to fabricate. Vehicle power/public-CAN interface only.

DRAFT — NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /08 / DELTA ABS GEN 2 MAIN CONNECTOR/

File: 08_ABS_Main.kicad_sch

Title: 08 / DELTA ABS GEN 2 MAIN CONNECTOR

Size: A3	Date: 2026-10-27	**Rev: J — REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 8/19

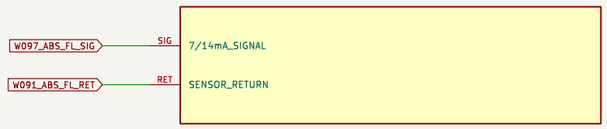
09 / ABS SENSORS, CONTROLS AND PUBLIC CAN

Wheel sensors -> Delta ABS -> public CAN -> R3 traction control. Selected path; Gen 2 protocol check remains open.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

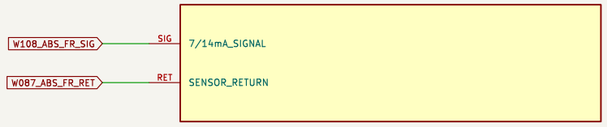
B20 | FRONT LEFT / GM 22742183

Front left wheel-speed sensor



B21 | FRONT RIGHT / GM 22742183

Front right wheel-speed sensor

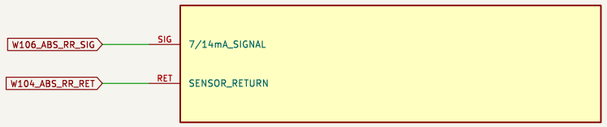


B22 | REAR LEFT / DELTA SENSOR

Rear left wheel-speed sensor

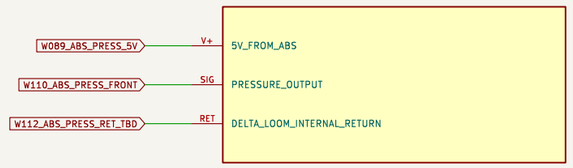
B23 | REAR RIGHT / DELTA SENSOR

Rear right wheel-speed sensor



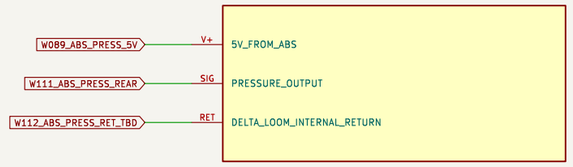
B24 | FRONT BRAKE PRESSURE

Front brake pressure sensor



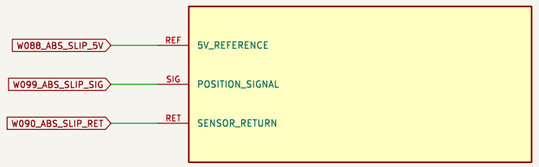
B25 | REAR BRAKE PRESSURE

Rear brake pressure sensor



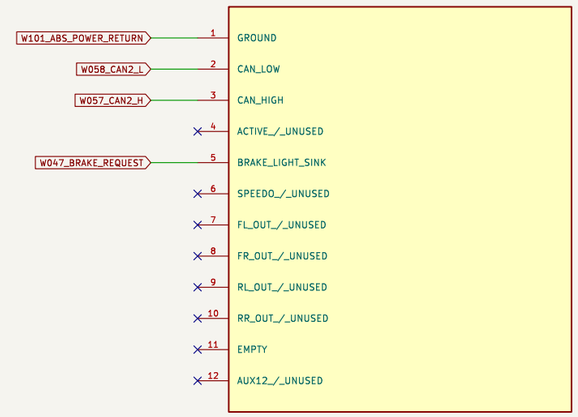
S20 | SLIP SELECTOR / PHYSICAL

Anti-lock brake slip selector



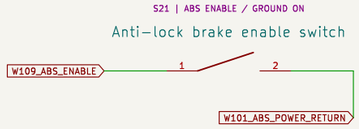
J22 | PUBLIC DTM06-12S

Anti-lock brake vehicle interface



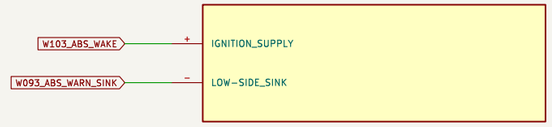
S21 | ABS ENABLE / GROUND ON

Anti-lock brake enable switch



H20 | ABS WARNING LAMP

Anti-lock brake warning lamp



HOLD A03: GM 22742183 signal compatibility/polarity unverified. Check both corners; GM lists it as front-left assembly.

Both pressure sensors use labelled 3-pin Delta loom plugs. Return W112 is supplier-internal; no added ground wire or R3 tap.

IMU/yaw sensor and its private CAN remain in supplied harness: map with verified harness data before build release.

S21/C158 HOLD: current guide describes hardware solenoid-power OFF; old pin35 ground-enable map is not a fabrication instruction.

CAN wheel speeds selected per Haltech support. J22:7-10 unused; verify Gen 2 profile/bitrate (sheet 15). SPI2 = brake request.

DRAFT - NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /09 / ABS SENSORS, CONTROLS AND PUBLIC CAN/

File: 09_ABS_Sensors.kicad_sch

Title: 09 / ABS SENSORS, CONTROLS AND PUBLIC CAN

Size: A3	Date: 2026-10-27	**Rev: J - REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 9/19

10 / CAN, DASH, KEYPAD AND WAKE

CAN1: R3 / PD16 / iC-7 / keypad / TMS-4 (1 Mbit/s Haltech). CAN2: Delta ABS public bus, separately verified.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

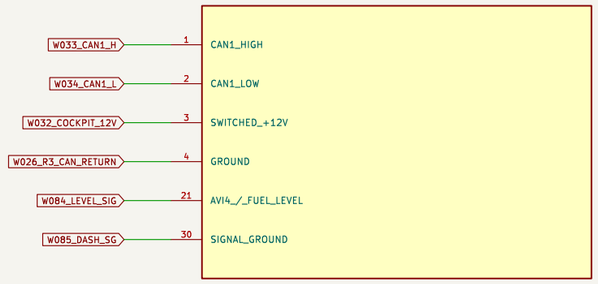
S1 | NORMAL R3 WAKE / U2 PERMIT

Engine controller power switch



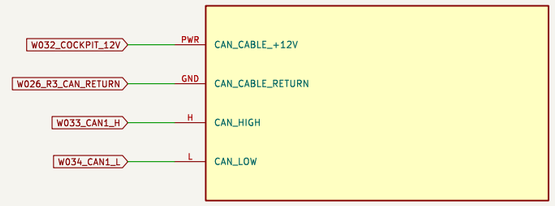
U10 | HALTECH IC-7

Haltech iC-7 digital dash



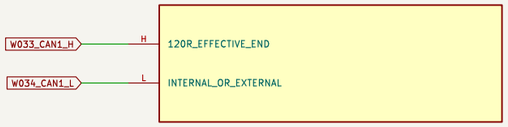
U11 | HT-011502 / 15 BUTTON

Driver control keypad



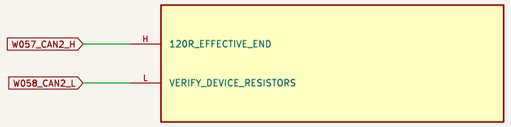
J30 | PD16 INTERNAL / CAN1 END

Dashboard data-bus termination



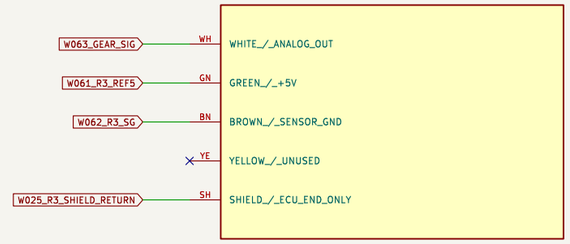
J31 | CAN2 FAR-END TERM

Vehicle data-bus termination



B30 | NOVOTECHNIK RSC-2841-634-211-202

Transaxle gear-position sensor



KEYPAD / CAMERA CONFIRMED; POSITION 10 PROPOSED

1 Run/Stop 2 Start (momentary) 3 Headlights 4 Tail lights 5 Hazards

6 Fan override 7 Timed fuel prime 8 Dash next 9 Alarm acknowledge 10 Camera ON/OFF

11 Spare 12 Spare 13 Spare 14 Spare 15 Spare

GEAR DISPLAY – R3 TO IC-7 OVER CAN1

NSP: Gear Position Sensor; AVI10 / C27; pull-up OFF.

Calibrate every forward gear, neutral and reverse in place.

Set non-overlapping voltage windows; check repeatability.

IC-7: display ECU Gear channel. No separate display needed.

S1 is normal wake, supplied only by CARTEK U2 M6_IGN. Emergency stop uses U2/S40 and R3 A13; verify engine/output shutdown and no backfeed.

CAN termination is a topology requirement, not two extra resistors to fit blindly. R3 has selectable internal termination per bus.

CAN1: R3 termination at one end; choose PD16 at the other and enable its termination. J30 represents that same resistor.

Twist each H/L pair; minimize stubs. Do not tie CAN1 to CAN2. R3 A11 total device return current must stay within its rating.

iC-7 fuel level: AVI4 pin 21 / return 30; 240R pull-up for resistive sender. Confirm dash hardware and NSP firmware.

DRAFT – NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /10 / CAN, DASH, KEYPAD AND WAKE/

File: 10_CAN_Cockpit.kicad_sch

Title: 10 / CAN, DASH, KEYPAD AND WAKE

Size: A3	Date: 2026-10-27	**Rev: J – REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 10/19

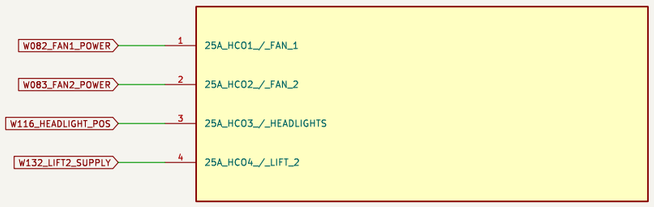
11 / AUXILIARY DISTRIBUTION AND LIGHTING

One Haltech PD16 / HT-198100 kit. U20, J40 (sheet 13) and J41 are interfaces on the SAME module.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J41 | PD16 B / DTP4

Haltech PD16 – 25 A power outputs



U20 | PD16 / HT-198100

Haltech PD16 – main power



Control, CAN and all four ground wires: J40 on sheet 13.

Main +12V: 4 AWG manufacturer baseline; F50 rating TBD.

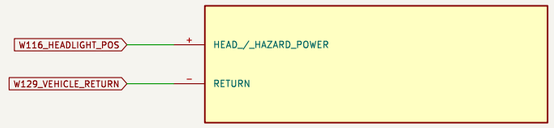
F50 | AUX MAIN / TBD

PD16 main supply fuse



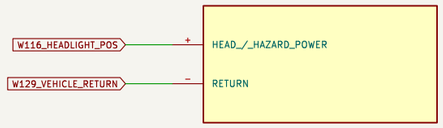
H30 | LEFT HEADLIGHT / TBD

Left LED headlight



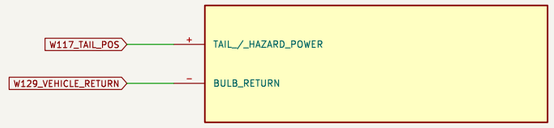
H37 | RIGHT HEADLIGHT / TBD

Right LED headlight



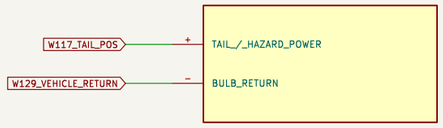
H31 | LEFT TAIL LED / TBD

Left tail LED bulb



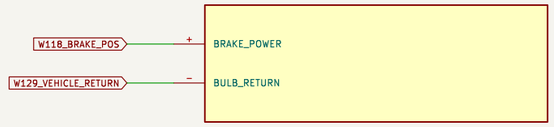
H35 | RIGHT TAIL LED / TBD

Right tail LED bulb



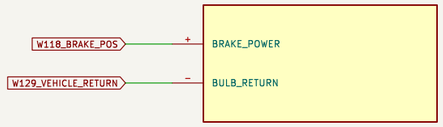
H34 | LEFT BRAKE LED / TBD

Left brake LED bulb



H36 | RIGHT BRAKE LED / TBD

Right brake LED bulb



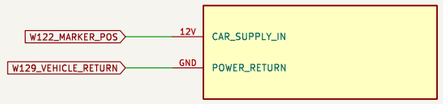
WOLFBOX CAMERA SYSTEM – SHEET 17

A7 / HCO5 -> kit -> USB-C mirror power.

Keypad camera toggle; proposed position 10.

U22 | BRAKE MARKER / CONVERTER

Brake-marker electronics



HEADLIGHT_POS = both headlights / PD16 B3. TAIL_POS = both tail bulbs / A3. BRAKE_POS = both brake bulbs / A4.

Keypad hazards: flash headlights AND tail bulbs together. Hazard logic overrides steady ON so the lamps can turn OFF each cycle.

Brake bulbs stay steady on brake request; no hazard flashing of brake bulbs. A5/HCO3 and A6/HCO4 are now SPARE.

Hazards OFF: restore normal head/tail requests. Implement shared flash phase and verify LED-driver cycling; no lamp models selected.

HOLD L01: configure/test flash logic, CAN-loss behavior, bulb currents, conductor/contact protection and repeated-start allowance.

DRAFT – NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /11 / AUXILIARY DISTRIBUTION AND LIGHTING/
File: 11_Auxiliary_Power.kicad_sch

Title: 11 / AUXILIARY DISTRIBUTION AND LIGHTING

Size: A3	Date: 2026-10-27	**Rev: J – REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 11/19

12 / HARNESS CONVENTIONS AND OPEN ITEMS

This is a connected functional draft for review. It is not a fabrication release or an ECU configuration file.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

HOW TO READ THIS SET

Identical Wxxx_NAME global labels are electrically connected across sheets.

Wxxx identifies an electrical net; branched nets require individual harness conductor IDs later.

Numeric R3/PD16/Delta/IC-7 pins are documented. Alphabetic functional IDs on TBD parts are not cavity numbers.

NC marks identify deliberately unused pins. A net ending TBD may have only one endpoint.

Reference J10/J11/J12 identifies R3 connectors A/C/E; J1 identifies its main power interface.

GROUNDING AND CONSTRUCTION

Battery/engine/chassis returns, R3 sensor return, R3 ground outputs and ABS sensor returns are distinct nets.

Do not bond R3 ground-output pins A10/A11 to chassis. Use A10 for shield drains at the ECU end only.

Reuse L94 donor plugs/pigtails after inspection; verify cavity identity and record color changes.

Wire sizes, insulation class, connector contacts, splice sizes and route lengths remain build-stage selections.

Electronic output protection must protect EVERY branch wire/contact; verify trip curves, cranking drop and startup.

RELEASE HOLDS

P01 CARTEK GT alone selected; verify engine-stop input, return isolation, cutoff timing, charging and starter driver.

E01 Crank/cam PNs identified; verify actual cavities, coil/injector/DBW hardware, 200mA reference and 500mA return budgets.

P02 Electronic trip curves vs weakest injector/coil/pump branches; F01 load/startup, fan settings and hot-idle test.

A01 ABS ground wording; A02 private/public CAN mapping; A03 C7 sensor signal/polarity compatibility.

N01 Gen 2 CAN profile/bitrate, TC loss behavior (sheet 15), PD16 setup; gear/HGT calibration and fuel-level input details.

CONFIGURATION AND COMMISSIONING

Load/program NSP separately: I/O mapping, output protection, sensor calibration, DBW faults and start permissives.

Check continuity/cavity orientation with electronics unplugged; then check supplies before connecting sensors.

Verify master cutoff with engine running, fuel stall shutdown, fan control and brake-light response.

Review source revisions and actual hardware before termination. No vehicle-powered test was performed.

NASA CCR is the requested safety reference; this drawing does not certify compliance.

Selected CAN traction-control path and spare I/O: sheet 15. Sources and pin/net schedule: README.md / Connections.tsv.

Verified manufacturer electrical limits are ceilings, not chosen fuse-trip settings. No fuse or wire purchase schedule is released.

DRAFT - NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /12 / HARNESS CONVENTIONS AND OPEN ITEMS/
File: 12_Design_Notes.kicad_sch

Title: 12 / HARNESS CONVENTIONS AND OPEN ITEMS

Size: A3	Date: 2026-10-27	**Rev: J - REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 12/19

13 / HALTECH PD16 CONNECTIONS AND CONTROL

Owner uses HT-188000 flying-lead harness | PD16 diagram rev 1.5, October 2024 | Peripheral Power Mode / Voltage Sense

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

J40 | PD16 A / AMP34 KEY1

Haltech PD16 – connector A

	1	HBO1_/_SPARE
	2	HBO2_/_SPARE
W117_TAIL_POS	3	8A_HCO1_/_TAIL_LIGHTS
W118_BRAKE_POS	4	8A_HCO2_/_BRAKE_LIGHTS
	5	8A_HCO3_/_SPARE
	6	8A_HCO4_/_SPARE
W121_CAMERA_POS	7	8A_HCO5_/_CAMERA
W122_MARKER_POS	8	8A_HCO6_/_BRAKE_MARKER
	9	8A_HCO7_/_SPARE
	10	8A_HCO8_/_SPARE
W124_PD16_SG	11	SENSOR_RETURN_/_OIL_+_TRANS
	12	5V_REFERENCE_/_UNUSED
	13	IGN_INPUT_/_UNUSED
	14	SPI1_/_SPARE
	15	SPI2_/_SPARE
	16	SPI3_/_SPARE
	17	8A_HCO9_/_SPARE
	18	8A_HCO10_/_SPARE
	19	SPI4_/_SPARE
W125_OIL_TEMP_SIG	20	AVI1_/_OIL_TEMP
W135_TRANS_TEMP_SIG	21	AVI2_/_TRANS_TEMP
W026_R3_CAN_RETURN	22	HALTECH_CAN_GROUND
W033_CAN1_H	23	HALTECH_CAN_HIGH
W034_CAN1_L	24	HALTECH_CAN_LOW
W131_PD16_WAKE	25	CAN_+12V_WAKE_INPUT
	26	IGN_SOURCE_/_UNUSED
	27	AVI3_/_SPARE
	28	AVI4_/_SPARE
	29	NO_CONNECTION
	30	NO_CONNECTION
W129_VEHICLE_RETURN	31	BATTERY_GROUND_1
W129_VEHICLE_RETURN	32	BATTERY_GROUND_2
W129_VEHICLE_RETURN	33	BATTERY_GROUND_3
W129_VEHICLE_RETURN	34	BATTERY_GROUND_4

ONE MODULE – THREE DRAWING REFERENCES

U20 = main positive; J41 = connector B; J40 = connector A.

Peripheral mode: dedicated R3 HBO6/A25 -> PD16 A25.

A13 and A26 are unused. Do not backfeed A25 as an output.

Ground A31–A34 to VEHICLE_RETURN, not BT1 negative.

Use four 18 AWG ground wires per Haltech guide.

A22 is CAN cable ground; not a substitute for A31–A34.

FAN CONFIGURATION – NO EXTERNAL PWM

Assign PD16 25A HCO1 / HCO2 to separate thermofans.

Use coolant-temperature on/off hysteresis and keypad override.

Temperature thresholds must match the thermostat/cooling test.

Verify power-up, CAN loss, sensor-fault and overcurrent behavior.

Do not make fan-off on track unconditional; retain overheat demand.

Measure hot-idle charging with both fans on; listed pair load 20 A.

CAN hub branch: H/L/return only. Keep PD16 A25 power separate from cockpit +12V; no four-wire power bridge.

Use the manufacturer connector-face view for cavity orientation. A/B prefixes identify connectors, not additional modules.

A3–A10/A17–A18 outputs are limited to 8 A each; B1–B4 to 25 A each. Main connector rating is not an aggregate thermal guarantee.

Configure Peripheral Power Mode, Haltech CAN 1 Mbit/s and PD16 end termination in NSP. Do not add another resistor at J30.

PD16 manufacturer baseline: main 4 AWG, B outputs 10 AWG, A high-side outputs 18 AWG. Final routing/derating/protection remain open.

HT-188000 supplies controller-side leads and CAN plug. Finish free ends; no cockpit-positive bridge into dedicated PD16 wake.

DRAFT – NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /13 / HALTECH PD16 CONNECTIONS AND CONTROL/
File: 13_PD16_Control.kicad_sch

Title: 13 / HALTECH PD16 CONNECTIONS AND CONTROL

Size: A3	Date: 2026-10-27	**Rev: J – REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 13/19

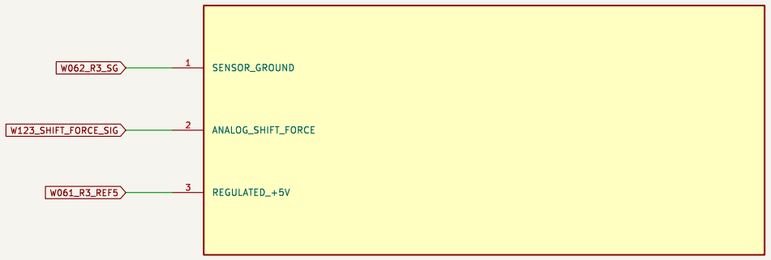
14 / HGT SHIFT KNOB AND FLAT–SHIFT CONTROL

B31 senses driver shift force. B30 (sheet 10) senses actual gear. Both signals are read directly by the Nexus R3.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

B31 | HGT STRAIN–GAUGED KNOB / DT STYLE

HGT shift–force sensing knob



R3 CONNECTIONS – CONNECTOR C / J11, SHEET 03

Knob cavity 1 –> C26 / sensor signal ground.

Knob cavity 2 –> C28 / AVI11; pull–up disabled.

Knob cavity 3 –> C25 / regulated sensor +5 V.

Internal amplifier: no external amplifier or cut relay.

Shared 5 V load budget: 200 mA maximum at C25.

HGT current demand is unlisted; verify before release.

HARNESS AND INSTALLATION

Proposed harness extension: three 22 AWG automotive wires; verify contact size and routing.

Use the supplied DT–style cavity numbers; verify the actual mating shell and terminal view.

Wire colors are unverified. Do not assign functions by color. Ground returns to C26 only.

Retain lead strain relief and a flexible service loop through full lever travel.

Mount for fore/aft force. M12 x 1.75 fitment; check adapter against the actual Weddle lever.

NSP SETUP – REQUIRED BEFORE ENABLING SHIFT CUT

Select Load Cell as flat–shift trigger; assign AVI11. Keep cut/blip disabled during initial checks.

At the selected 5 V supply, HGT resting output is about 1.80 V. Measure rest, pull and push.

Confirm actual signal direction; calibrate trigger voltages and hysteresis for driver force.

Use B30 / AVI10 calibrated gear feedback for closed–loop shift setup; validate every gear.

Tune RPM/TPS enable limits, cut/recovery, maximum cut time and retrigger blockout.

Verify sensor–disconnect behavior. Auto–blip remains disabled until DBW/downshift tuning.

Source: HGT installation guide 63800–IB–SCK–008, 12 July 2024, especially page 5; see README for links.

Pinout applies to HGT pre–terminated DT–style units. Confirm supplied hardware before terminating the mating harness.

HOLD S01: supply–current budget, mating connector, installed signal calibration and tuner validation. No NSP file programmed.

DRAFT – NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /14 / HGT SHIFT KNOB AND FLAT–SHIFT CONTROL/
File: 14_Shift_Control.kicad_sch

Title: 14 / HGT SHIFT KNOB AND FLAT–SHIFT CONTROL

Size: A3	Date: 2026–10–27	**Rev: J – REVIEW**
KiCad E.D.A. kicad–cli 7.0.10		Id: 14/19

15 / TRACTION CONTROL AND SPARE I/O

Selected: Delta CAN wheel speeds to R3. Haltech support response supplied by owner; see README / Sources.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

SELECTED WHEEL-SPEED PATH

Four wheel sensors -> Delta ABS (sheets 08-09).

Delta public J22:3 CAN H / J22:2 CAN L -> R3 CAN2.

R3 J11/C21 = CAN2 H; J11/C22 = CAN2 L (sheet 03).

NSP: select the available Delta ABS vehicle-CAN profile.

Assign FL, FR, RL and RR speed sources to CAN input.

R3 compares calibrated front/rear speeds for traction control.

J22:7-10 pulse outputs stay NC; R3 SPI3-6 stay spare.

Sensor signal/return pairs remain connected to Delta only.

SUPPORT CONFIRMATION AND CHECKS

Haltech support: CAN is suitable; Delta profile is available.

Gen 2 qualification: verify protocol against installed NSP.

Confirm public-bus bitrate, termination and harness mapping.

Published wheel-speed broadcast: 100Hz / nominal 10ms.

Check wheel identity, units and calibration for all four wheels.

Validate TC enable, slip tuning and sensor/CAN fault response.

Do not infer receive/TC rate from ECU-to-dash broadcasts.

This drawing does not configure NSP or validate live data.

SPARE CHANNELS - REVISION J

R3: 1 dedicated AVI; 4 SPI; 6 DPO; 0 HBO; 0 HCO.

R3 AVI9/C18 spare (onboard MAP); SPI3-6 remain spare.

Fuel pressure AVI6/C15; intake air temperature AVI8/C17.

PD16: 2 AVI; 4 SPI; 2 HBO; 6 x 8A HCO; 0 x 25A HCO.

PD16 AVI1/A20 engine oil; AVI2/A21 trans; A11 return.

iC-7: 3 spare legacy sensor AVI1-3; AVI4 = fuel level.

NSP additionally exposes AVI5-9, 3 DPI and 4 DPO.

Extended dash capacity is firmware-dependent; see README.

S50 | HT-010504 / 12-POSITION

Traction-control level selector



S50:1 -> C25 5V; S50:2 -> C5/SPI1; S50:3 -> C26.

SPI1 analog rotary input; input pull-up MUST be OFF.

Use HT-010504 calibration matching the fitted decal.

TC levels/slip/cut/fault behavior remain tuner setup.

No wheel-sensor split; Delta CAN2 speed path unchanged.

NSP: PD16 AVI1 = Engine Oil Temperature; AVI2 = Transmission Oil Temperature. Separate HT-010306 calibrations.

iC-7 fuel level: pin 21, sensor return pin 30. NSP sharing minimums: R3 1.26.4 / dash 2.26.3; verify installed versions.

TMS-4 uses CAN1 via supplied adapter (sheet 16). CAN2 remains Delta only; keep two effective terminations per bus.

HOLD T01: Delta Gen 2 profile/bitrate, wheel calibration and traction-control failure behavior before enabling TC.

HOLD F02: combined R3 5V reference current, cockpit supply/return load and camera-kit variant/protection test.

DRAFT - NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /15 / TRACTION CONTROL AND SPARE I/O/

File: 15_Traction_IO.kicad_sch

Title: 15 / TRACTION CONTROL AND SPARE I/O

Size: A3	Date: 2026-10-27	**Rev: J - REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 15/19

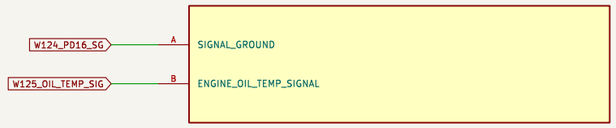
16 / TMS-4 TPMS, ENGINE AND TRANSMISSION OIL TEMP

Two HT-010306 high-range 150C sensors use separate PD16 AVIs. HT-011600 tyre sensors communicate wirelessly.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

B32 | HT-010306 / ENGINE OIL

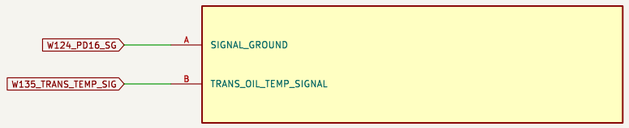
HT-010306 engine oil temperature



Engine: PD16 A20 / AVI1 -> B32:B; A11 -> B32:A.

B33 | HT-010306 / TRANS OIL

HT-010306 transmission oil temperature



Transmission: PD16 A21 / AVI2 -> B33:B; A11 -> B33:A.

Enable each AVI internal 1k pull-up; no external 5V wire.

Both: load Haltech Oil 150C 1/8 NPT HT010306.cal.

Separate engine/trans channels; verify curve and sharing.

EV1 harness view: A return / B signal. A11: sensors only.

FOUR WHEEL SENSORS / WIRELESS

Fit internal sensors to FL / FR / RL / RR; batteries are inside the sensors. No wires run out to the wheels.

NSP: Connections > CAN > Haltech CAN System > Tyre Monitoring / TMS-4 Enable; assign IDs to actual corners.

TMS-4: verify corner identity, pressure/temperature units, alarm thresholds and lost-sensor response. Mount receiver high in vehicle.

Manufacturer typical in-motion refresh: 60 seconds. TPMS is for tyre monitoring; it is NOT a wheel-speed source for traction control.

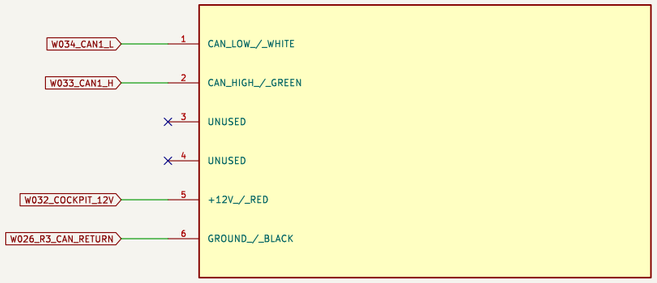
PD16 engine/trans temp: separate open/short diagnostics, valid range, alarms and stale-CAN behavior. Thresholds remain unset.

Confirm trans mounting/immersion, thread clearance, 150C range and calibration against a reference; no mechanical location specified.

Sources: Haltech TMS-4 QSG; HT-010306 info sheet rev 4.2; PD16 diagram rev 1.5. See Sources/ and README.

U23 | HT-011600 / TMS-4 MCON6

Haltech TMS-4 receiver / HT-011600



Supplied DTM4 -> MCON6 adapter:

DTM1 +12V -> MCON5; DTM2 GND -> MCON6.

DTM3 CAN H -> MCON2; DTM4 CAN L -> MCON1.

Receiver: 12-32V; 50mA ambient operating current.

Use supplied adapter; do not infer plug viewing direction.

DRAFT - NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /16 / TMS-4 TPMS, ENGINE AND TRANSMISSION OIL TEMP/
File: 16_TPMS_Oil_Temp.kicad_sch

Title: 16 / TMS-4 TPMS, ENGINE AND TRANSMISSION OIL TEMP

Size: A3	Date: 2026-10-27	**Rev: J - REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 16/19

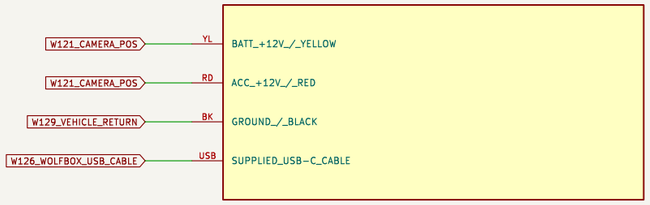
17 / WOLFBOX G900 PRO CAMERA POWER

Keypad request –> R3 CAN control –> PD16 A7 HCO5. Complete power removal when commanded OFF; no parking supply.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

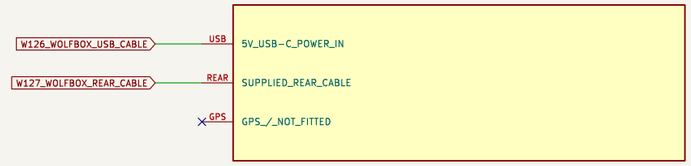
U24 | G900 PRO HARDWIRE KIT

WolfBox G900 Pro hardwire converter



U21 | G900 PRO MIRROR / FRONT CAM

WolfBox G900 Pro mirror and front camera



SWITCHED-ONLY DESIGN / VERIFY KIT ON RECEIPT

Yellow BATT AND red ACC take the same camera-switched

12V feed from PD16 A7. Black returns to vehicle return.

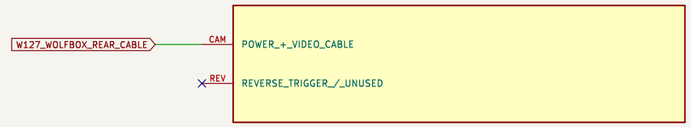
This deliberately removes all camera power when OFF.

Retain the WolfBox converter and its supplied USB-C cable.

Use a kit matching the mirror version/input-current label.

U25 | WOLFBOX REAR CAMERA

WolfBox rear camera



No separate 12V or USB supply goes to the rear camera.

Select normal rear view; disable screen auto-off as needed.

CONTROL AND PROTECTION

Camera button toggles PD16 A7; proposed keypad position 10, initial OFF. R3 / PD16 / keypad must already be awake.

Use high-side ON/OFF, full supply when ON; do not PWM the converter supply. Verify orderly recording on power loss.

USB and rear-cable W labels denote complete vendor cable assemblies, NOT single wires or USB cavity numbers.

HOLD C01: confirm purchased kit labels/variant and switched-only operation with both BATT/ACC on A7 before installation.

3A/4A at 5V are mirror supply ratings, not measured continuous draw. At 12V / 90% efficiency: about 1.4A / 1.9A input.

Select PD16 trip/inrush/retry settings to protect actual kit leads; retain vendor inline protection if fitted. Do not use 8A blindly.

No raw-battery parking feed; camera stays downstream of master isolation. GPS and reverse lead omitted as the simple rear-view basis.

Insulate unused reverse lead. Verify boot time, rear view, CAN-loss OFF, cranking and master-stop behavior on actual hardware.

DRAFT – NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /17 / WOLFBOX G900 PRO CAMERA POWER/

File: 17_Wolfbox_Camera.kicad_sch

Title: 17 / WOLFBOX G900 PRO CAMERA POWER

Size: A3	Date: 2026-10-27	**Rev: J – REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 17/19

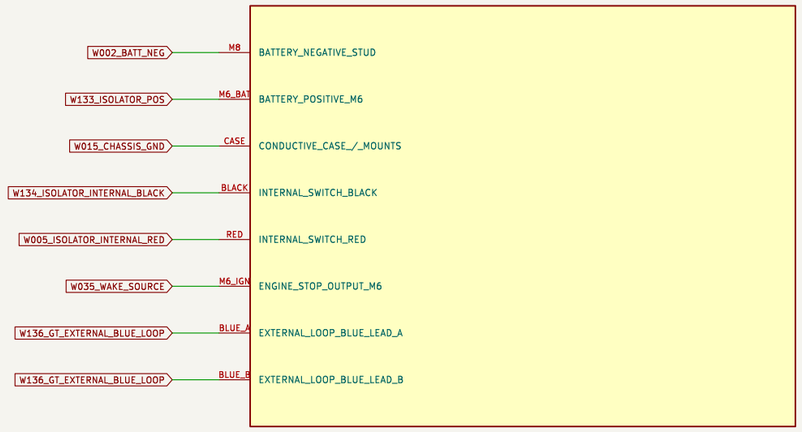
18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS

GT manual rev1.0 / 07Nov2025, method 1 ECU/PDM input. ONE internal stop switch; no external switch. Blocks are proposed parts.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

U2 | CARTEK GT / VERIFY s OR x

CARTEK GT battery isolator



S40 | B0ONTT91Y0 / LATCHING NC

BAOMAIN latching stop / NC



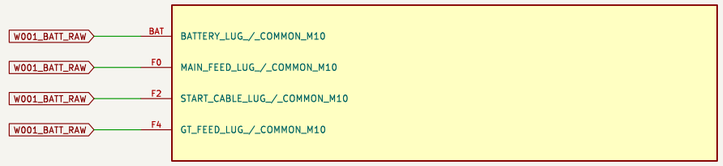
F4 | GT CONTROL / 5A PROPOSED

GT CONTROL / 5A PROPOSED



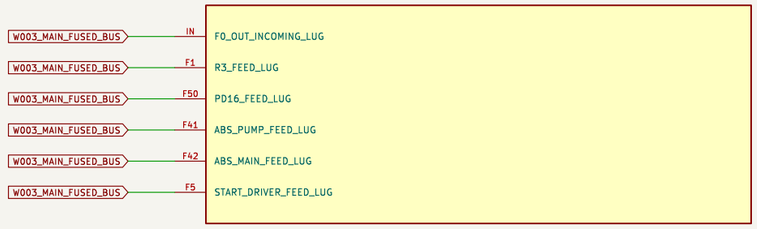
TB1 | FS46540 / RAW + / PROPOSED

Sierra FS46540 / PROPOSED



TB2 | SUM-900533 / FUSED + / PROPOSED

Summit SUM-900533 / PROPOSED



S40: one cockpit stop button, reachable inside/outside.

Reset/released = NC closed / ON; pressed = latched open / OFF.

Verify actual NC pair. Pins 1/2 are functional, NOT cavity IDs.

Join BLUE_A to BLUE_B permanently; no chassis connection. S41 removed. No battery voltage on switch leads.

GT M6_IGN -> S1 -> R3 A13 only. K1 removed; F0 -> TB2 direct. GT output is fixed 40A protected.

TB1: four lugs / common M10 post, stack and covers TBD. TB2: six lugs / four 3/8 studs; allocation TBD.

TB2 listing conflicts: specific 250A vs family 150A. Confirm actual rating, thermal duty and permitted lug stacks.

GT: 7-18V; 600A cranking / 2000A surge. Confirm received GT-s/GT-x, terminal labels and actual starter current.

Install in manual order: chassis bond, switch/blue loop, engine-stop, M6 battery positive, then M8 negative. Limit temp to +85C pending source conflict.

F4 5A remains PROPOSED for the GT/control feed. Size wiring/protection for actual GT + R3 A13 load, not a contactor coil.

K1 deleted by owner. Verify R3 A13 OFF stops engine/fuel and PD16 outputs promptly before GT battery isolation.

Twist reset may re-arm power; require a fresh start command. Charger/booster cables both at battery posts; no negative bypass.

DRAFT – NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS/

File: 18_Master_Stop.kicad_sch

Title: 18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS

Size: A3	Date: 2026-10-27	**Rev: J – REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 18/19

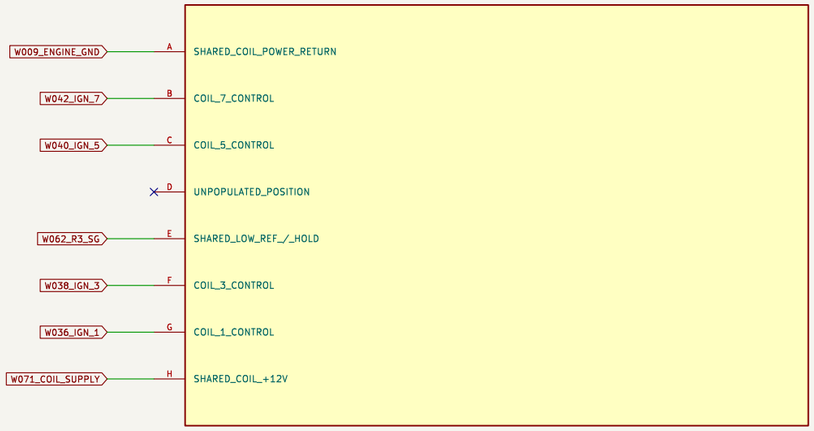
19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS

Retained donor wiring / 7 populated bank ways / related-year GM cavity reference; confirm actual donor continuity and orientation.

REV J | DESIGN REVIEW DRAFT | TBD = NOT RELEASED FOR WIRING

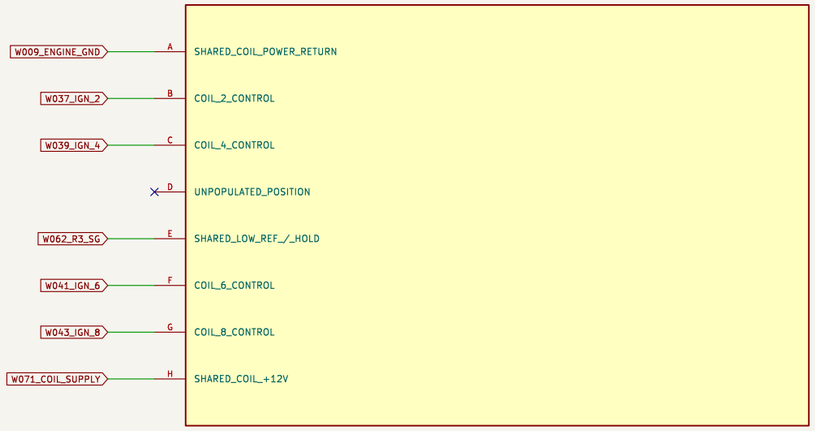
X111 | OEM L94 / ODD / 7-WAY

Stock L94 odd-bank coil connector



X113 | OEM L94 / EVEN / 7-WAY

Stock L94 even-bank coil connector



OEM COIL BANK LOOMS — RETAIN THEIR INTERNAL SPLICES

Each bank: four triggers plus common +12V, power ground and low reference. D is unavailable, not an eighth live wire.

Reference coil plug: A=power ground, B=low reference, C=trigger, D=+12V. Device IDs on sheet 04 stay functional.

S paths in the continuous views are RETAINED STOCK wiring. C paths are new integration wires; no new coil-branch cuts assumed.

DONOR INJECTOR WIRING — REUSE WITH LS3 EV6 / USCAR INJECTORS

Retain eight individual driver legs and separate odd/even +12V branches. No factory injector bulkhead plug is assumed.

CUT-INJ labels mark functional donor cut/termination boundaries, NOT connector cavities; locations/join hardware TBD.

Factory feed circuits may also serve coils: isolate retained injector feeds before HCO1. NEVER connect HCO1 to HCO2.

HOLD: verify LS3 injector PN, plug key/latch, connector orientation, harness reach, condition, continuity and calibration.

GROUND HOLD: bank A -> engine; bank E -> R3 C26 is retained pending exact coil evidence. No W009/W062 bridge permitted.

GM separates low reference; Haltech generic Gen IV guide also advises both coil grounds at the head. Resolve with exact coil/R3 guidance.

Reference: Stock_Engine_Harnesses.json and OEM_Harness_Interfaces.tsv. Source cavities/colors are not inspected donor measurements.

No downstream bank fuses restored. HCO1 injector / HCO2 coil trip coordination with every OEM branch and contact remains OPEN.

DRAFT — NOT A BUILD RELEASE

Danny's Race Car | Harness design

Sheet: /19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS/
File: 19_Stock_Engine_Harnesses.kicad_sch

Title: 19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS

Size: A3	Date: 2026-10-27	**Rev: J — REVIEW**
KiCad E.D.A. kicad-cli 7.0.10		Id: 19/19